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Swingset

Abstract

A swingset includes at least two support frames distributed at intervals and a connecting rod assembly configured to connect the at least two support frames. The connecting rod assembly has a length direction and includes a first connecting rod, a second connecting rod, and a reinforcing member. The first connecting rod and the second connecting rod are arranged in sequence along the length direction. The reinforcing member overlaps a joint of the first connecting rod and the second connecting rod, and the reinforcing member is detachably connected to the first connecting rod and the second connecting rod. The reinforcing member is configured to disperse force at the joint of the first connecting rod and the second connecting rod during operation of the swingset, and weaken tension at the joint during operation of the swingset.

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Background/Summary

FIELD
(1) The subject matter herein generally relates to playground equipments, for example, a swingset.
BACKGROUND
(2) Traditional swingset has a long beam, which is not conducive to packaging and transportation, and can be easily bent and deformed during transportation and use, resulting in a shortened service life of the swingset.
SUMMARY
(3) What is needed, is a swingset.
(4) A swingset includes at least two support frames distributed at intervals and a connecting rod assembly configured to connect the at least two support frames. Each of the at least two support frames include at least one leg. The connecting rod assembly has a length direction and includes a first connecting rod, a second connecting rod, and a reinforcing member. The first connecting rod and the second connecting rod are arranged in sequence along the length direction. The reinforcing member overlaps a joint of the first connecting rod and the second connecting rod, and the reinforcing member is detachably connected to the first connecting rod and the second connecting

rod. The reinforcing member is configured to disperse force at the joint of the first connecting rod and the second connecting rod during operation of the swingset, and weaken tension at the joint during operation of the swingset, so that the service life of the swingset is prolonged.
(5) A swingset includes two support frames distributed at intervals and a cross rod assembly detachably connected to the support frames. The cross rod assembly includes a connecting rod assembly, the connecting rod assembly has a length direction and includes a first connecting rod, a second connecting rod, a reinforcing member, and at least one fastener. The first connecting rod and the second connecting rod are arranged in sequence along the length direction. The reinforcing member overlaps a joint of the first connecting rod and the second connecting rod, the at least one fastener detachably fixes the reinforcing member, the first connecting rod, and the second connecting rod together. In a cross-section perpendicular to the length direction, the reinforcing member overlaps at least one third of a perimeter of the first connecting rod, and the reinforcing member overlaps at least one third of a perimeter of the second connecting rod. Each of the support frames includes at least one leg, each of the at least one leg is connected to the cross rod assembly. The first connecting rod and the second connecting rod are detachably connected, which is conducive to the packaging and transportation of the swingset, and is conducive to reducing the probability of bending and deformation caused by a long length of the connecting rod during transportation. Furthermore, the reinforcing member is arranged at the joint of the first connecting rod and the second connecting rod, and the first connecting rod, the second connecting rod, and the reinforcing member are fixed by the at least one fastener, so that the joint of the first connecting rod and the second connecting rod is not easy to bending, which is beneficial to ensure the safety of the swingset and prolong the service life of the swingset.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Implementations of the present disclosure will now be described, by way of embodiments, with reference to the attached figures.
(2) FIG. **1** is a diagram of an embodiment of a connecting rod assembly according to the present disclosure.
(3) FIG. **2** is a partially exploded, diagrammatic view of an embodiment of a connecting rod assembly according to the present disclosure.
(4) FIG. **3**A is a partially exploded, diagrammatic view of an embodiment of a connecting rod assembly according to the present disclosure.
(5) FIG. **3**B is a partially exploded, diagrammatic view of an embodiment of a connecting rod assembly according to the present disclosure.
(6) FIG. **4** is a partially exploded, diagrammatic view of an embodiment of a connecting rod assembly according to the present disclosure.
(7) FIG. **5** is a diagram of an embodiment of a connecting rod assembly according to the present disclosure.
(8) FIG. **6** is a diagram of an embodiment of a connecting rod assembly according to the present disclosure.
(9) FIG. **7**A is a partially exploded, diagrammatic view of an embodiment of a connecting rod assembly according to the present disclosure.
(10) FIG. **7**B is a partially exploded, diagrammatic view of an embodiment of a connecting rod assembly according to the present disclosure.
(11) FIG. **7**C is a partially exploded, diagrammatic view of an embodiment of a connecting rod assembly according to the present disclosure.
(12) FIG. **8** is a partially exploded, diagrammatic view of an embodiment of a connecting rod

assembly according to the present disclosure.
(13) FIG. **9** is a partially exploded, diagrammatic view of an embodiment of a connecting rod assembly according to the present disclosure.
(14) FIG. **10** is a partially exploded, diagrammatic view of an embodiment of a connecting rod assembly according to the present disclosure.
(15) FIG. **11**A is a cross-sectional view of an embodiment of a connecting rod assembly according to the present disclosure.
(16) FIG. **11**B is a cross-sectional view of an embodiment of a connecting rod assembly according to the present disclosure.
(17) FIG. **12**A is a diagram of an embodiment of a swingset according to the present disclosure.
(18) FIG. **12**B is a partially diagram of an embodiment of a swingset according to the present disclosure.
(19) FIG. **12**C is a partially diagram of an embodiment of a swingset according to the present disclosure.
(20) FIG. **13** is a diagram of an embodiment of a swingset according to the present disclosure.
(21) FIG. **14** is a partially diagram of an embodiment of a swingset according to the present disclosure.
(22) FIG. **15** is a partially diagram of an embodiment of a swingset according to the present disclosure.
(23) FIG. **16** is a diagram of an embodiment of a swingset according to the present disclosure.
(24) FIG. **17** is a diagram of an embodiment of a swingset according to the present disclosure.
(25) FIG. **18**A is a diagram of an embodiment of a swingset according to the present disclosure.
(26) FIG. **18**B is a diagram of an embodiment of a swingset according to the present disclosure.
(27) FIG. **19** is a diagram of an embodiment of a swingset according to the present disclosure.
(28) FIG. **20** is a diagram of an embodiment of a swingset according to the present disclosure.
(29) FIG. **21** is a diagram of an embodiment of a swingset according to the present disclosure.

DETAILED DESCRIPTION

(30) It will be appreciated that for simplicity and clarity of illustration, where appropriate, reference numerals have been repeated among the different figures to indicate corresponding or analogous elements. In addition, numerous specific details are set forth in order to provide a thorough understanding of the embodiments described herein. However, it will be understood by those of ordinary skill in the art that the embodiments described herein can be practiced without these specific details. In other instances, methods, procedures, and components have not been described in detail so as not to obscure the related relevant feature being described. Also, the description is not to be considered as limiting the scope of the embodiments described herein. The drawings are not necessarily to scale, and the proportions of certain parts may be exaggerated to better illustrate details and features of the present disclosure.
(31) The term “comprising,” when utilized, means “including, but not necessarily limited to”; it specifically indicates open-ended inclusion or membership in the so-described combination, group, series, and the like.
(32) FIG. **1** and FIG. **2** illustrate an embodiment of a connecting rod assembly **10**. The connecting rod assembly **10** is applicable in a piece of playground equipment, for example, in a swingset. The connecting rod assembly **10** includes a first connecting rod **11** and a second connecting rod **12** arranged in sequence along a length direction X of the connecting rod assembly **10**. The connecting rod assembly **10** further includes a reinforcing member **13** and at least one fastener **14**. The reinforcing member **13** is overlaped at a joint A of the first connecting rod **11** and the second connecting rod **12**. The at least one fastener **14** detachably fixes the reinforcing member **13**, the first connecting rod **11**, and the second connecting rod **12**. The first connecting rod **11** and the second connecting rod **12** are detachably connected, which can reduce the volume of whole assembly at packing for advantageous product packing and shipping and cost reduction for clearing

export customs, and can reduce the probability of bending and deformation caused by a long length of the connecting rod during transportation. Furthermore, the reinforcing member **13** is arranged at the joint A of the first connecting rod **11** and the second connecting rod **12**. Then the first connecting rod **11**, the second connecting rod **12**, and the reinforcing member **13** are fixed by the at least one fastener **14**, so that the joint A of the first connecting rod **11** and the second connecting rod **12** is not easy to be bent, which is beneficial to ensure the safety of the product and prolong the service life of the product.

(33) For example, when the connecting rod assembly **10** is applicable in a swingset, the first connecting rod **11** and the second connecting rod **12** are assembled into a cross rod assembly of the swingset. The reinforcing member **13** is arranged at the joint A of the first connecting rod **11** and the second connecting rod **12**. Then, the reinforcing member **13**, the first connecting rod **11**, and the second connecting rod **12** are detachably fixed by the at least one fastener **14**. So that the stability and the bearing capacity of the cross rod assembly of the swingset can be enhanced. For example, when an attachment (e.g., a swing, or a climbing strap) is suspended on the cross rod assembly, the force at the joint A can be dispersed by the reinforcing member **13**, so that the joint A is not easily bent or even broken. For another example, a first swing is suspended on one side of the joint A of the cross rod assembly and a second swing is suspended on the other side of the joint A, and the first swing swings forward along a direction perpendicular to the length direction X and the second swing swings backward along a direction perpendicular to the length direction X. On the one hand, the force at the joint A can be dispersed by the reinforcing member **13**, and on the other hand, a tension at the joint A can be weakened by the reinforcing member **13**. So that the swingset does not easily to shake or even bent, or even broken. As a result, the service life of the swingset is prolonged.

(34) In at least one embodiment, the tension at the joint A includes at least one of a first tension, a second tension, and a third tension. The first tension is generated between the first connecting rod **11** and the second connecting rod **12**. The second tension is generated between the first connecting rod **11** and the reinforcing member **13**. and the third tension is generated between the second connecting rod **12** and the reinforcing member **13**.

(35) In at least one embodiment, the first connecting rod **11** may be in a shape of a hollow tube and includes a first end **11***a* and a second end **11***b* facing away from the first end **11***a*. The second connecting rod **12** may be in a shape of a hollow tube and includes a first end **12***a* and a second end **12***b* facing away from the first end **12***a*. The second end **11***b* of the first connecting rod **11** is connected to the first end **12***a* of the second connecting rod **12** to form the joint A. When a shape of the first connecting rod **11** is the same as a shape of the second connecting rod **12**, and a size of the first connecting rod **11** is the same as a size of the second connecting rod **12**, an end surface of the second end **11***b* of the first connecting rod **11** and an end surface of the first end **12***a* of the second connecting rod **12** are aligned and connected. The end surface of the second end **11***b* of the first connecting rod **11** and the end surface of the first end **12***a* of the second connecting rod **12** may be in complete contact at the joint A, or a little gap may be existed at the joint A. It should be noted that, in actual use, the smaller the gap, the better, and the smaller the gap, the stronger the stability of the connecting rod assembly.

(36) When the shape and the size of the first connecting rod **11** are different from the shape and the size of the second connecting rod **12**, the second end **11***b* of the first connecting rod **11** and the first end **12***a* of the second connecting rod **12** are sleeved, and the joint A includes an overlapping area of the first connecting rod **11** and the second connecting rod **12**. For example, if the second end **11***b* of the first connecting rod **11** is sleeved on the first end **12***a* of the second connecting rod **12**, and the first end **12***a* of the second connecting rod **12** is received in the first connecting rod **11**. The reinforcing member **13** is overlaped at the second end **11***b* of the first connecting rod **11**. Correspondingly, if the first end **12***a* of the second connecting rod **12** is sleeved on the second end **11***b* of the first connecting rod **11**, and the second end **11***b* of the first connecting rod **11** is received

in the second connecting rod **12**. The reinforcing member **13** is overlapped at the first end **12***a* of the second connecting rod **12**.
(37) In at least one embodiment, the first connecting rod **11** may be cylindrical, the second connecting rod **12** may be cylindrical, and a diameter of the end surface of the first connecting rod **11** may be the same as a diameter of the end surface of the second connecting rod **12**. In at least one embodiment, the first connecting rod **11** and the second connecting rod **12** may also have other shapes, such as but not limited to elliptical cylinders and rectangular parallelepipeds. The size of the end surface of the first connecting rod **11** and the size of the end surface of the second connecting rod **12** may be the same or may be different. Preferably, the shape of the first connecting rod **11** is consistent with the shape of the second connecting rod **12**, the size of the first connecting rod **11** is consistent with the size of the second connecting rod **12**. In the present disclosure, the following description will be given by taking the end surface of the cylindrical first connecting rod **11** and the end surface of the cylindrical second connecting rod **12** having the same size as an example.
(38) Preferably, in at least one embodiment, the first connecting rod **11** may be arched, the second connecting rod **12** may also be arched. When both of the first connecting rod **11** and the second connecting rod **12** are arched, the first connecting rod **11** and the second connecting rod **12** protrude toward the same side.
(39) In at least one embodiment, referring to FIGS. **1** and **2**, the reinforcing member **13** includes two reinforcing sheets, which are a first reinforcing sheet **131** and a second reinforcing sheet **133** respectively. Wherein, the first reinforcing sheet **131** is an arc-shaped sheet, and the second reinforcing sheet **133** is an arc-shaped sheet.
(40) The first reinforcing sheet **131** is in contact with (e.g., attached to) an outer surface of the first connecting rod **11** and an outer surface of the second connecting rod **12**, the second reinforcing sheet **133** is attached to the outer surface of the first connecting rod **11** and the outer surface of the second connecting rod **12**. The first reinforcing sheet **131** and the second reinforcing sheet **133** are arranged opposite to each other. For example, when the connecting rod assembly **10** is applicable in a swingset, the first connecting rod **11** and the second connecting rod **12** are assembled into a cross rod assembly of the swingset. In practical use the first reinforcing sheet **131** is mounted on a bottom surface of the cross rod assembly, and the second reinforcing sheet **133** is mounted on a top surface of the cross rod assembly. Or the first reinforcing sheet **131** is mounted on a top surface of the cross rod assembly, and the second reinforcing sheet **133** is mounted on a bottom surface of the cross rod assembly.
(41) The first reinforcing sheet **131**, the first connecting rod **11**, and the second connecting rod **12** may have the same contour. Therefore, the first reinforcing sheet **131** is sized and shaped to overlap at least part of correspondingly shaped first connecting rod **11** and at least part of correspondingly shaped second connecting rod **12**.
(42) The second reinforcing sheet **133**, the first connecting rod **11**, and the second connecting rod **12** may have the same contour. Therefore, the second reinforcing sheet **133** is sized and shaped to overlap at least part of correspondingly shaped first connecting rod **11** and at least part of correspondingly shaped second connecting rod **12**.
(43) It can be understood that when the first connecting rod **11** and the second connecting rod **12** are in many other kinds of sizes and shapes, a shape and size of the first reinforcing sheet **131** and a shape and size of the second reinforcing sheet **133** also change accordingly.
(44) In at least one embodiment, the at least one fastener **14** may include a plurality of bolts, which are at least one first bolt **141** and at least one second bolt **143**. Correspondingly, at least a pair of first mounting holes **110** are oppositely arranged on the first connecting rod **11**, at least a pair of second mounting holes **120** are oppositely arranged on the second connecting rod **12**, at least two third mounting holes **132** are arranged on the first reinforcing sheet **131**, and at least two fourth mounting holes **134** are arranged on the second reinforcing sheet **133**. Each first bolt **141** extends

through one of the third mounting holes **132**, two opposite first mounting holes **110**, and one of the fourth mounting holes **134** in sequence, and fixes and locks the first reinforcing sheet **131**, the first connecting rod **11**, and the second reinforcing sheet **133** together. Each second bolt **143** extends through one of the third mounting holes **132**, two opposite second mounting holes **120**, and one of the fourth mounting holes **134** in sequence, and fixes and locks the first reinforcing sheet **131**, the second connecting rod **12**, and the second reinforcing sheet **133** together. Specifically, in at least one embodiment, the number of the first bolts **141** is two, the number of the second bolts **143** is two, the numbers of the first mounting holes **110**, the second mounting holes **120**, the third mounting holes **132**, and the fourth mounting holes **134** are vary accordingly. In at least one embodiment, the first reinforcing sheet **131** and the second reinforcing sheet **133** are arranged oppositely at the joint A. Two first bolts **141** and two second bolts **143** are arranged in sequence along the length direction X, so that the connecting rod assembly **10** is not easily to be bent at the joint A after being stressed.

(45) The connecting rod assembly **10** may further include a connecting member, at least one of the first connecting rod **11** and the second connecting rod **12** is sleeved on the connecting member. That is, the connecting member is received in at least one of the first connecting rod **11** and the second connecting rod **12**, so that when the connecting rod assembly **10** is stressed, the connecting member can disperse force at the joint A of the first connecting rod **11** and the second connecting rod **12**, thereby preventing the joint A of the first connecting rod **11** and the second connecting rod **12** from being easily deformed, bent or broken. On the other hand, when the connecting rod assembly **10** is applicable in a swingset, during the operation of the swingset, a lateral tension generated by the swingset can be weakened, thereby preventing the swingset from easily shaking. In at least one embodiment, the connecting member may be a reinforcing member. Preferably, along the length direction X, a length of the reinforcing member **13** is greater than or equal to a distribution length of mounting holes on the connecting member. In at least one embodiment, the length of the reinforcing member **13** may be greater than or equal to a length of the connecting member.

(46) In at least one embodiment, referring to FIG. **3**A, the connecting member includes an extension portion **121** extending along the length direction X from one end of the second connecting rod **12**. In one embodiment, the extension portion **121** and the second connecting rod **12** are integrally formed. A cross section of the extension portion **121** perpendicular to the length direction X is less than a cross section of the second connecting rod **12** perpendicular to the length direction X, that is, a stepped surface **122** is formed at a joint of the extension portion **121** and the second connecting rod **12**. The first connecting rod **11** is in a shape of a hollow tube and is sleeved on the extension portion **121**. And the first connecting rod **11** may abut against the stepped surface **122**. Then, at least a pair of fifth mounting holes **123** are oppositely arranged on the extension portion **121**, and the first bolt **141** further extends through two opposite fifth mounting holes **123**. In at least one embodiment, the second bolts **143** and the second mounting holes **120** may be omitted, the first connecting rod **11** and the second connecting rod **12** are fixed together by the first bolt **141** extending through the third mounting hole **132**, the first mounting holes **110**, the fifth mounting holes **123**, and the fourth mounting holes **134**.

(47) Along the length direction X, a length of the first reinforcing member **131** is greater than a distribution length of the fifth mounting holes **123** on the extension portion **121**, and a length of the second reinforcing member **133** is greater than the distribution length of the fifth mounting holes **123** on the extension portion **121**. So that the structural stability of the connecting rod assembly **10** is enhanced.

(48) In at least one embodiment, along the length direction X, the length of the first reinforcing member **131** can be equal to the distribution length of the fifth mounting holes **123** on the extension portion **121**, and the length of the second reinforcing member **133** can be equal to the distribution length of the fifth mounting holes **123** on the extension portion **121**.

(49) In at least one embodiment, along the length direction X, the length of the first reinforcing sheet **131** is greater than or equal to the length of the extension portion **121**, the length of the second reinforcing sheet **133** is greater than or equal to the length of the extension portion **121**. Preferably, along the length direction X, the length of the first reinforcing sheet **131** is twice the length of the extension portion **121**, the length of the second reinforcing sheet **133** is twice the length of the extension portion **121**. Half length of the first reinforcing sheet **131** overlaps the extension portion **121**, and the other half length of the first reinforcing sheet **131** overlaps the second connecting rod **12**. Half length of the second reinforcing sheet **133** overlaps the extension portion **121**, and the other half length of the second reinforcing sheet **133** overlaps the second connecting rod **12**. So that the structural stability of the connecting rod assembly **10** is further enhanced.
(50) FIG. **3**B shows a connecting rod assembly different from the connecting rod assembly of the embodiment shown in FIG. **3**A. Referring to FIG. **3**B, the connecting rod assembly includes a first bolt **141** and a second bolt **143**. Specifically, the first bolt **141** and the second bolt **143** are arranged at intervals along the length direction X. The first bolt **141** extends through one third mounting hole **132**, one of the first mounting holes **110**, two opposite fifth mounting holes **123**, the other of the first mounting holes **110**, and one fourth mounting hole **134** in sequence. The second bolt **143** extends through one third mounting hole **132**, two opposite second mounting holes **120**, and one fourth mounting hole **134** in sequence.
(51) In at least one embodiment, referring to FIG. **4**, the connecting member includes a connecting pillar **15**. The connecting pillar **15** includes a first end **151** and a second end **153** facing away from the first end **151**. The first connecting rod **11** and the second connecting rod **12** are respectively in a shape of a hollow tube. Wherein, the first connecting rod **11** is sleeved on the connecting pillar **15** from the first end **151**, and the second connecting rod **12** is sleeved on the connecting pillar **15** from the second end **153**. The connecting pillar **15** is received in the first connecting rod **11** and the second connecting rod **12**. And the first connecting rod **11** and the second connecting rod **12** may abut against with each other. Then, at least two pairs of sixth mounting holes **150** are arranged on the connecting pillar **15**. The first bolt **141** further extends through one pair of the sixth mounting holes **150**, the second bolt **143** further extends through the other pair of the sixth mounting holes **150**.
(52) Along the length direction X, the length of the first reinforcing sheet **131** is greater than or equal to a length of the connecting pillar **15**, the length of the second reinforcing sheet **133** is greater than or equal to a length of the connecting pillar **15**. So that the structural stability of the connecting rod assembly **10** is enhanced. Preferably, along the length direction X, the length of the first reinforcing sheet **131** is equal to the length of the connecting pillar **15**, and opposite ends of the first reinforcing sheet **131** are flush with opposite ends of the connecting pillar **15**. For example, the first reinforcing sheet **131** includes a first end and a second end facing away from the first end. The connecting pillar **15** includes a first end and a second end facing away from the first end. The first end of the first reinforcing sheet **131** is flush with the first end of the connecting pillar **15**, and the second end of the first reinforcing sheet **131** is flush with the second end of the connecting pillar **15**. Along the length direction X, the length of the second reinforcing sheet **133** is equal to the length of the connecting pillar **15**, and opposite ends of the second reinforcing sheet **133** are flush with opposite ends of the connecting pillar **15**. For example, the second reinforcing sheet **133** includes a third end and a fourth end facing away from the third end. The third end of the second reinforcing sheet **133** is flush with the first end of the connecting pillar **15**, and the fourth end of the second reinforcing sheet **133** is flush with the second end of the connecting pillar **15**. So that the structural stability of the connecting rod assembly **10** is improved.
(53) It can be understood that, in at least one embodiment, the length of the first reinforcing sheet **131** or the length of the second reinforcing sheet **133** may be slightly less than the length of the connecting pillar **15**. Along the length direction X, the length of the first reinforcing sheet **131** is

greater than or equal to a distribution length of sixth mounting holes **150** on the connecting pillar **15**, and the length of the second reinforcing sheet **133** is greater than or equal to a distribution length of sixth mounting holes **150** on the connecting pillar **15**. For example, referring to FIG. **4**, the length d**1** of the first reinforcing sheet **131** is greater than or equal to the distribution length d**2** of sixth mounting holes **150** on the connecting pillar **15**, then d**1**custom characterd**2**.
(54) In at least one embodiment, referring to FIGS. **2** and **5**, the second reinforcing sheet **133** may be omitted, only the first reinforcing sheet **131** is attached to the outer surface of the first connecting rod **11** and the outer surface of the second connecting rod **12**, each first bolt **141** extends through the first mounting holes **110** and the third mounting hole **132** to fix and lock the first connecting rod **11** and the first reinforcing sheet **131** together, and each second bolt **143** extends through the second mounting holes **120** and another third mounting hole **132** to fix and lock the second connecting rod **12** and the first reinforcing sheet **131** together, thereby realizing a purpose of fixing and locking the first connecting rod **11** and the second connecting rod **12**. And the first connecting rod **11** may abut with the second connecting rod **12**.
(55) In at least one embodiment, referring to FIGS. **6** and **7**A, the reinforcing member **13** may include a sleeve, and the sleeve is sleeved on the outer surface of the first connecting rod **11** and the outer surface of the second connecting rod **12**. The second end **11***b* of the first connecting rod **11** and the first end **12***a* of the second connecting rod **12** are received in the sleeve. At least two pairs of seventh mounting holes **130** are oppositely arranged on the sleeve. Wherein, each first bolt **141** extends through two opposite seventh mounting holes **130** and two opposite first mounting holes **110**, so the first connecting rod **11** and the sleeve are fixed and locked together. Each second bolt **143** extends through another two opposite seventh mounting holes **130** and two opposite second mounting holes **120**, and the second connecting rod **12** and the sleeve are fixed and locked together. And the first connecting rod **11** may abut against the second connecting rod **12**.
(56) FIG. **7**B shows a connecting rod assembly different from the connecting rod assembly of the embodiment shown in FIG. **7**A. Referring to FIG. **7**B, the connecting rod assembly further includes a connecting member, the connecting member includes an extension portion **121** extending along the length direction X from one end of the second connecting rod **12**. In one embodiment, the extension portion **121** and the second connecting rod **12** are integrally formed. A cross section of the extension portion **121** perpendicular to the length direction X is less than a cross section of the second connecting rod **12** perpendicular to the length direction X, that is, a stepped surface **122** is formed at a joint of the extension portion **121** and the second connecting rod **12**. The first connecting rod **11** is in a shape of a hollow tube and is sleeved on the extension portion **121**. And the first connecting rod **11** may abut against the stepped surface **122**. Then, at least a pair of fifth mounting holes **123** are oppositely arranged on the extension portion **121**, and the first bolt **141** further extends through two opposite fifth mounting holes **123**. In at least one embodiment, the second bolts **143** and the second mounting holes **120** may be omitted, the first connecting rod **11** and the second connecting rod **12** are fixed together by the first bolt **141** extending through two opposite seventh mounting holes **130**, two opposite first mounting holes **110**, and the fifth mounting holes **123**.
(57) Along the length direction X, a length of the reinforcing member **13** is greater than a distribution length of the fifth mounting holes **123** on the extension portion **121**. So that the structural stability of the connecting rod assembly **10** is enhanced.
(58) In at least one embodiment, along the length direction X, the length of the reinforcing member **13** can be equal to the distribution length of the fifth mounting holes **123** on the extension portion **121**.
(59) In at least one embodiment, along the length direction X, the length of the reinforcing member **13** is greater than or equal to the length of the extension portion **121**. Preferably, along the length direction X, the length of the reinforcing member **13** is twice the length of the extension portion **121**. Half length of the reinforcing member **13** overlaps the extension portion **121**, and the other

half length of the reinforcing member **13** overlaps the second connecting rod **12**. So that the structural stability of the connecting rod assembly **10** is further enhanced. In at least one embodiment, along the length direction X, the length of the reinforcing member **13** is greater than or equal to a distribution length of the fifth mounting holes **123** of the extension portion **121** as shown in FIG. **7**B.

(60) FIG. **7**C shows a connecting rod assembly different from the connecting rod assembly of the embodiment shown in FIG. **7**B. Referring to FIG. **7**C, the connecting member includes a connecting pillar **15**. The connecting pillar **15** includes a first end **151** and a second end **153** facing away from the first end **151**. The first connecting rod **11** and the second connecting rod **12** are respectively in a shape of a hollow tube. Wherein, the first connecting rod **11** is sleeved on the connecting pillar **15** from the first end **151**, and the second connecting rod **12** is sleeved on the connecting pillar **15** from the second end **153**. The connecting pillar **15** is received in the first connecting rod **11** and the second connecting rod **12**. And the first connecting rod **11** and the second connecting rod **12** may abut against with each other. Then, at least two pairs of sixth mounting holes **150** are arranged on the connecting pillar **15**. The first bolt **141** further extends through one pair of the sixth mounting holes **150**, the second bolt **143** further extends through the other pair of the sixth mounting holes **150**.

(61) Along the length direction X, the length of the reinforcing member **13** is greater than or equal to a length of the connecting pillar **15**. So that the structural stability of the connecting rod assembly **10** is enhanced. Preferably, along the length direction X, the length of the reinforcing member **13** is equal to the length of the connecting pillar **15**, and opposite ends of the reinforcing member **13** are flush with opposite ends of the connecting pillar **15**. For example, the reinforcing member **13** includes a first end and a second end facing away from the first end. The connecting pillar **15** includes a first end and a second end facing away from the first end. The first end of the reinforcing member **13** is flush with the first end of the connecting pillar **15**, and the second end of the reinforcing member **13** is flush with the second end of the connecting pillar **15**. So that the structural stability of the connecting rod assembly **10** is improved.

(62) It can be understood that, in at least one embodiment, the length of the reinforcing member **13** may be slightly less than the length of the connecting pillar **15**. Along the length direction X, the length of the reinforcing member **13** is greater than or equal to a distribution length of sixth mounting holes **150** on the connecting pillar **15**. For example, referring to IFG. **7**C, the length d**3** of the reinforcing member **13** is greater than or equal to the distribution length d**2** of sixth mounting holes **150** on the connecting pillar **15**, then d**3**custom characterd**2**.

(63) In at least one embodiment, referring to FIGS. **8** and **11**B, the sleeve as the reinforcing member **13** may include a slit **135**. The sleeve further includes two opposite ends, an inner surface, and an outer surface facing away from the inner surface. The slit **135** extends from one of the ends of the sleeve toward the other end, and penetrates the inner surface and the outer surface of the sleeve. Referring to FIG. **9**, the slit **135** may penetrate the two opposite ends, and penetrate the inner surface and the outer surface of the sleeve. Referring to FIG. **10**, when the slit **135** extends from one of the ends of the sleeve toward the other end of the sleeve, the number of the slits **135** may be multiple and multiple slits **135** may distribute at intervals.

(64) In combination with the above-mentioned embodiments that the reinforcing member **13** may be a reinforcing sheet, two reinforcing sheet or a sleeve, in at least one embodiment, referring to FIGS. **11**A and **11**B, in a cross-section perpendicular to the length direction X, in an overlapping area of the reinforcing member **13** and the first connecting rod **11**, the reinforcing member **13** overlaps at least one third of a perimeter of the first connecting rod **11**. And in an overlapping area of the reinforcing member **13** and the second connecting rod **12**, the reinforcing member **13** overlaps at least one third of a perimeter of the second connecting rod **12**, so as to help improve a strength of the joint of the first connecting rod **11** and the second connecting rod **12**, so that the joint is not easy to be bent, and the structure of the connecting rod assembly **10** is more stable.

More preferably, in the cross-section perpendicular to the length direction X, in the overlapping area of the reinforcing member **13** and the first connecting rod **11**, the reinforcing member **13** overlaps at least half of the perimeter of the first connecting rod **11**, and in the overlapping area of the reinforcing member **13** and the second connecting rod **12**, the reinforcing member **13** overlaps at least half of the perimeter of the second connecting rod **12**. In at least one embodiment, in a cross-section of an element, a perimeter of the element refers to the perimeter of the outer surface of the element in the cross-section.
(65) In at least one embodiment, along the length direction X, an overlapping length of the reinforcing member **13** and the first connecting rod **11** may be equal to an overlapping length of the reinforcing member **13** and the second connecting rod **12**.
(66) In at least one embodiment, the connecting rod assembly **10** may be formed by several connecting rods which are sequentially connected, and every two connecting rods are connected by the reinforcing members and fasteners.
(67) In at least one embodiment, if the connecting rod assembly **10** includes the connecting member, the reinforcing member **13** may be omitted.
(68) Referring to FIG. **12**A, the above-mentioned connecting rod assembly **10** may be applicable in a swingset **100**. The swingset **100** includes a cross rod assembly **10***a* and at least two support frames **30**. The at least two support frames **30** are distributed at intervals. The cross rod assembly **10***a* is connected between the at least two support frames **30**. In the present disclosure, the number of the support frames **30** is two, and the cross rod assembly **10***a* is detachably connected to each of the support frames **30** as an example for the following description.
(69) The cross rod assembly **10***a* includes the connecting rod assembly **10** as described above. The at least two support frames **30** are arranged at intervals along the length direction X, and the connecting rod assembly **10** connects to the at least two support frames **30**. In at least one embodiment, the connecting rod assembly **10** is detachably connected to the at least two support frames **30**. In some embodiment, the connecting rod assembly **10** may be directly fixedly connected to the at least two support frames **30** by welding.
(70) Each of the at least two support frames **30** includes at least one leg **31**. In this embodiment, each support frame **30** preferably include two legs **31**. In at least one embodiment, the legs **31** are arranged obliquely. Each of the support frames **30** may further include a lateral support member **32**, the lateral support member **32** is connected between the two legs **31**. A triangular structure is formed between the two legs **31** and the lateral support member **32**, so that a structure of each of the support frames **30** is stable. Each of the support frames **30** may further include a reinforcing support member **34**, one end of the reinforcing support member **34** is connected to the lateral support member **32**, the other end extends away from the cross rod assembly **10***a*. Specifically, the end of the reinforcing support member **34** away from the lateral support member **32** is used to support against the ground. The reinforcing support member **34** is used in combination with the lateral support member **32** to form the swingset **100** can prevent the swingset **100** from shaking in the length direction X, so that a structure of the swingset **100** is more stable.
(71) Each of the legs **31** is connected to the cross rod assembly **10***a*. Specifically, in at least one embodiment, the cross rod assembly **10***a* may be provided with two leg sleeves **101** spaced from each other along the length direction X. Each of the leg sleeves **101** includes two intersecting connecting portions **102** respectively sleeved on the legs **31** of the support frame **30**, so as to connect the support frame **30** and the cross rod assembly **10***a*. Referring to FIG. **12**A, the leg sleeves **101** are disposed adjacent to opposite ends of the cross rod assembly **10***a*. In at least one embodiment, the leg sleeve **101** may be disposed at any position of the cross rod assembly **10***a*, and the number of the leg sleeves **101** is adjusted according to the number of the support frames **30**.
(72) In at least one embodiment, referring to FIG. **12**B, the swingset **100** may further include a strengthening member **13***a* and a fastening member **14***a*. The strengthening member **13***a* may be disposed at a joint of the connecting portion **102** and the leg **31**. In at least one embodiment, the

strengthening member **13***a* may be disposed at an overlapping area of the connecting portion **102** and the leg **31**, and the fastening member **14***a* detachably fixes the strengthening member **13***a*, the connecting portion **102**, and the leg **31**, thereby increasing a connection strength between the connecting portion **102** and the leg **31**, which is beneficial to prevent the joint of the connecting portion **102** and the leg **31** from being bent, even prevent the connecting portion **102** or the leg **31** from being broken due to a tension generated when the swing swings along the direction perpendicular to the length direction X. So that the structure of the swingset **100** is more stable during use, and the service life of the swingset **100** is prolonged. The fastening member **14***a* may include at least one bolt, and the number of the bolts is not limited. Specifically, ways to connect the connecting portion **102** to the leg **31** may be the same as the ways to connect the first connecting rod **11** to the second connecting rod **12**, and a fixing manner between the connecting portion **102** and the leg **31** may be the same as a fixing manner between the first connecting rod **11** and the second connecting rod **12**. A structure of the strengthening member **13***a* may be the same as a structure of the reinforcing member **13**, and a structure of the fastening member **14***a* may be the same as a structure of fastener **14**. In at least one embodiment, referring to FIG. **12***c*, the swingset **100** may further include a mounting pillar **15***a*, the mounting pillar **15***a* includes a first end portion **151***a* and a second end portion **153***a* facing away from the first end portion **151***a*. The connecting portion **102** is sleeved on the mounting pillar **15***a* from the first end portion **151***a*. The leg **31** is a hollow tube and is sleeved on the mounting pillar **15***a* from the second end portion **153***a*. The strengthening member **13***a* is disposed on the connecting portion **102** and the leg **31** and corresponds to the mounting pillar **15***a*. For example, a length of the strengthening member **13***a* is equal to a length of the mounting pillar **15***a*. Two opposite ends of the strengthening member **13***a* are aligned with two opposite ends of the mounting pillar **15***a*. In other embodiment, the length of the strengthening member **13***a* may be greater than the length of the mounting pillar **15***a*. The fastening member **14***a* detachably fixes the strengthening member **13***a*, the connecting portion **102**, the mounting pillar **15***a*, and the leg **31**.

(73) In at least one embodiment, referring to FIG. **13**, the swingset **100** may further include two strengthening support members **40**. One end of each of the strengthening support members **40** is connected to the cross rod assembly **10***a*, the other end of each of the strengthening support members **40** is connected to one of the support frames **30**. Specifically, each of the strengthening support members **40** is connected to the leg **31** of the support frames **30**. A shape of the strengthening support members **40** is not limited and varied as needed.

(74) Referring to FIG. **12**A, an end portion of each of the legs **31** facing away from the cross rod assembly **10***a* may be provided with a ground gripping member **33**. Specifically, in at least one embodiment, referring to FIG. **14**, the ground gripping member **33** may include a ground support portion **331** and a fixing member **333**. The ground support portion **331** is used to support on the ground and is substantially parallel to the cross rod assembly **10***a* and the fixing member **333** is matched with the ground support portion **331**. The ground support portion **331** includes a connecting hole **332**, the fixing member **333** movably extends through the connecting hole **332** to fix the support frame **30** to the ground. The fixing member **333** may be, but not limited to, a nail, such as an expansion screw or a ground nail. Referring to FIG. **15**, one end of the ground gripping member **33** may be fixed with the leg **31**, and the other end of the ground gripping member **33** is used for grasping the ground. For example, the other end of the ground gripping member **33** is inserted into the ground.

(75) An end of the reinforcing support member **34** facing away from the lateral support member **32** may include the ground gripping member **33** too, so that the swingset **100** is more stable during use.

(76) In at least one embodiment, the leg **31** may include the connecting rod assembly **10**, so that the leg **31** can be disassembled during transportation, a risk of the leg **31** being deformed due to being too long during transportation is reduced, and the firmness of a structure of the leg **31** may be

improved, thereby helping to ensure the safety of the swingset and prolong the service life of the swingset.
(77) The cross rod assembly **10***a* may include at least one hanger **16** (shown in FIG. **12**A). The at least one hanger **16** is arranged on the first connecting rod **11** or on the second connecting rod **12** for hanging objects. The number of the hangers **16** can be varied as needed. Referring to FIGS. **12**A and **16**, the swingset **100** includes at least one swing **41**, the swing **41** may be attached to the cross rod assembly **10***a* through the at least one hanger **16**. In at least one embodiment, the at least one hanger **16** may be omitted, the swing **41** can be attached to the cross rod assembly **10***a* through a hanging rope of the swing **41**.
(78) In at least one embodiment, referring to FIG. **17**, the swingset **100** may further include a basketball hoop **43**, and the basketball hoop **43** is arranged at an end of the cross rod assembly **10***a*. In at least one embodiment, the basketball hoop **43** may be arranged on the support frame **30**.
(79) The swingset **100** may further include a slide **45**, the slide **45** is arranged adjacent to the support frame **30** and is connected to the support frame **30**.
(80) In at least one embodiment, referring to FIG. **18**A, the swingset **100** may further include a climbing structure **47**, and an end of the climbing structure **47** is attached to the cross rod assembly **10***a*. In at least one embodiment, the cross rod assembly **10***a* attached to the climbing structure **47** may be other structures, not limited to the structures described in the foregoing content. For example, the cross rod assembly **10***a* may be a crossbar or other commonly structures used in the art.
(81) The number of the climbing structures **47** can be varied as needed, which is not limited in the present disclosure.
(82) The support frame **30** may be used as a part of a structure of the climbing structure **47**. Referring to FIGS. **18**A and **18**B, the climbing structure **47** includes a climbing assembly **471** connected between two legs **31**. The climbing assembly **471** may be a soft rope braided ladder, a steel cable braided ladder, or a horizontal ladder that is not easily deformed, and may also be a trapezoidal structure, a mesh structure, or other structures made of any other material. In at least one embodiment, the climbing structure **47** may be arranged on only one of the legs **31**. For example, referring to FIGS. **18**B and **19**, a plurality of protrusions distributed at intervals can be arranged on the leg **31** for climbing. A plurality of annular structures distributed at intervals may be arranged on the leg **31** for climbing.
(83) Referring to FIG. **20**, the climbing structure **47** includes the climbing assembly **471** and at least one support leg **472**. The climbing assembly **471** is disposed between one of the legs **31** and one of the at least one support leg **472** or is connected between two support legs **472**. In at least one embodiment, the climbing assembly **471** may be arranged on only one of the at least one support leg **472**.
(84) The support leg **472** may include the above connecting rod assembly **10**, so that the support leg **472** can be disassembled during transportation, a risk of the support leg **472** being deformed due to being too long during transportation is reduced, and the firmness of a structure of the support leg **472** may be improved, thereby helping to ensure the safety of the swingset and prolong the service life of the swingset.
(85) Referring to FIG. **21**, the climbing structure **47** may be attached to the cross rod assembly **10***a*. The climbing structure **47** may be a rope ladder. The climbing structure **47** may be a mesh structure woven by ropes attached to the cross rod assembly **10***a*, and even more, the climbing structure **47** may be a rope with a plurality of spaced knots attached to the cross rod assembly **10***a*.
(86) It should be noted that the specific structure of the climbing structure **47** described in the present disclosure is not limited, as long as it can be pedaled and grasped for climbing.
(87) In addition, other recreational structures can be combined with the swingset **100**. For example, the other recreational structures may include a playhouse, climbing walls and the like.
(88) In at least one embodiment, the connecting rod assembly **10** and the swingset having the

connecting rod assembly **10** described in the present disclosure, the first connecting rod **11** and the second connecting rod **12** are detachably connected, which is conducive to the packaging and transportation of products, and is conducive to reducing the probability of bending and deformation caused by a long length of the connecting rod during transportation. Furthermore, the reinforcing member **13** is arranged at the joint A of the first connecting rod **11** and the second connecting rod **12**, and the at least one fastener **14** is used to fix the first connecting rod **11**, the second connecting rod **12**, and the reinforcing member **13**. so that the joint A of the first connecting rod **11** and the second connecting rod **12** is not easy to be bent, which is beneficial to ensure the safety of the product and prolong the service life of the product. In addition, the production process of the connecting rod assembly **10** is simple, which may improve the production efficiency of the product. Moreover, the reinforcing member **13** can be very stably fixed at the joint of the first connecting rod **11** and the second connecting rod **12** by the at least one fastener **14**, so that the product has a stable structure and is easy to be assembled.
(89) It is to be understood, even though information and advantages of the present embodiments have been set forth in the foregoing description, together with details of the structures and functions of the present embodiments, the disclosure is illustrative only; changes may be made in detail, especially in matters of shape, size, and arrangement of parts within the principles of the present embodiments to the full extent indicated by the plain meaning of the terms in which the appended claims are expressed.

Claims

1. A swingset comprising: at least two support frames distributed at intervals, each of the at least two support frames comprising at least one leg; and a connecting rod assembly configured to connect the at least two support frames, the connecting rod assembly having a length direction and comprising: a first connecting rod; a second connecting rod, the first connecting rod and the second connecting rod arranged in sequence along the length direction; and a reinforcing member overlapping a joint of the first connecting rod and the second connecting rod, and the reinforcing member detachably connected to the first connecting rod and the second connecting rod, wherein the reinforcing member is configured to disperse force at the joint of the first connecting rod and the second connecting rod during operation of the swingset, and weaken tension at the joint during operation of the swingset.
2. The swingset of claim 1, wherein the reinforcing member is received in the first connecting rod and the second connecting rod.
3. The swingset of claim 1, wherein the reinforcing member is mounted on an outer surface of the first connecting rod and an outer surface of the second connecting rod.
4. The swingset of claim 3, wherein the connecting rod assembly further comprises a connecting member, at least one of the first connecting rod and the second connecting rod is sleeved on the connecting member.
5. The swingset of claim 4, wherein at least two mounting holes are defined in the connecting member, and the at least two mounting holes have a distribution length along the length direction; at least one fastener extends through the mounting holes and detachably connects the connecting member, the reinforcing member, the first connecting rod, and the second connecting rod together along the length direction, a length of the reinforcing member is greater than or equal to the distribution length of the mounting holes on the connecting member.
6. The swingset of claim 1, wherein in a cross-section perpendicular to the length direction, the reinforcing member overlaps at least one third of a perimeter of the first connecting rod, and the reinforcing member overlaps at least one third of a perimeter of the second connecting rod.
7. The swingset of claim 1, wherein the reinforcing member comprises a reinforcing sheet, the reinforcing sheet is in contact with the first connecting rod and the second connecting rod

respectively.
8. The swingset of claim 1, wherein the reinforcing member comprises two reinforcing sheets, each of the reinforcing sheets is in contact with the first connecting rod and the second connecting rod respectively, the reinforcing sheets are arranged oppositely, at least one fastener further fixes the first connecting rod, the second connecting rod, and the reinforcing sheets together.
9. The swingset of claim 1, wherein the reinforcing member comprises a sleeve, and the sleeve is sleeved on the first connecting rod and the second connecting rod.
10. The swingset of claim 9, wherein the sleeve comprises a slit, opposite ends, an inner surface connecting the opposite ends, and an outer surface facing away from the inner surface, the slit extends from one of the ends of the sleeve toward the other end of the sleeve and penetrates the inner surface of the sleeve and the outer surface of the sleeve.
11. The swingset of claim 4, wherein the connecting member extends along the length direction from an end of the second connecting rod, the connecting member and the second connecting rod are integrally formed, an end of the first connecting rod is sleeve on the connecting member, the reinforcing member is correspondingly positioned to the connecting member.
12. A swingset comprising: at least two support frames distributed at intervals, each of the at least two support frames comprising at least one leg; and a cross rod assembly detachably connected to the support frames and comprising a connecting rod assembly, the connecting rod assembly having a length direction and comprising: a first connecting rod; a second connecting rod; a reinforcing member; and at least one fastener; wherein the first connecting rod and the second connecting rod are arranged in sequence along the length direction, the reinforcing member overlaps a joint of the first connecting rod and the second connecting rod, the at least one fastener detachably fixes the reinforcing member, the first connecting rod, and the second connecting rod together; in a cross-section perpendicular to the length direction, the reinforcing member overlaps at least one third of a perimeter of the first connecting rod, and the reinforcing member overlaps at least one third of a perimeter of the second connecting rod; each of the at least one leg of the support frames is connected to the cross rod assembly.
13. The swingset of claim 12, wherein the reinforcing member is mounted on an outer surface of the first connecting rod and an outer surface of the second connecting rod, the connecting rod assembly further comprises a connecting member, at least one of the first connecting rod and the second connecting rod is sleeved on the connecting member.
14. The swingset of claim 13, wherein at least two mounting holes are defined in the connecting member, and the at least two mounting holes have a distribution length along the length direction; the at least one fastener extends through the mounting holes and connects the reinforcing member, the first connecting rod, and the second connecting rod together; along the length direction, a length of the reinforcing member is greater than or equal to the distribution length of the mounting holes on the connecting member.
15. The swingset of claim 12, wherein the reinforcing member comprises a reinforcing sheet, the reinforcing sheet is in contact with the first connecting rod and the second connecting rod respectively.
16. The swingset of claim 12, wherein the reinforcing member comprises two reinforcing sheets, each of the reinforcing sheets is in contact with the first connecting rod and the second connecting rod respectively, the reinforcing sheets are arranged oppositely, the at least one fastener further fixes the first connecting rod, the second connecting rod, and the reinforcing sheets together.
17. The swingset of claim 12, wherein the reinforcing member comprises a sleeve, and the sleeve is sleeved on the first connecting rod and the second connecting rod.
18. The swingset of claim 17, wherein the sleeve comprises a slit, opposite ends, an inner surface connecting the opposite ends, and an outer surface facing away from the inner surface, the slit extends from one of the ends of the sleeve toward the other end of the sleeve and penetrates the inner surface of the sleeve and the outer surface of the sleeve.

19. The swingset of claim 13, wherein the connecting member extends along the length direction from an end of the second connecting rod, the connecting member and the second connecting rod are integrally formed, an end of the first connecting rod is sleeve on the connecting member, the reinforcing member is correspondingly positioned to the connecting member.

20. The swingset of claim 11, wherein each of the at least one leg comprises the connecting rod assembly.

21. The swingset of claim 11, wherein each of the support frames comprises two legs and further comprises a lateral support member and a reinforcing support member, the lateral support member is connected between the legs, and one end of the reinforcing support member is connected to the lateral support member, the other end extends away from the cross rod assembly.